

WATERWATCH · DESIGN DECISION RECORD

WaterWatch TRL 2 review decisions

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DOCUMENT CONTROL

Document ID	WWT-DDR-001	Version	0.2
Status	Draft	Date	2026-09-25
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/waterwatch		
Source	docs/decisions/0001-trl2-review-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	Record Amish's decisions on the TRL 2 review items and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** accepted for items D1 to D10; items O1 to O3 remain proposed

■ Context

The TRL 2 review note (docs/REVIEW.md , session 2026-09-25, /populate) listed ten items as "Proposed, awaiting Amish", and the design precis WWT-PRC-001 v0.2 listed key design choices, each with options and a recommendation. On 2026-09-25 Amish reviewed the review points for every portfolio repo and wrote: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them." Every item that carried a recommendation is therefore decided in favor of that recommendation. Items without a recommendation stay open. The same instruction approved portfolio-wide decisions on the SwapCell interface (v0.3), on pricing shared SwapCell packs once, and on community designs picking co-design partners per area later. Only the last applies to WaterWatch.

■ Options considered

The options for each item are those listed in docs/REVIEW.md (session 2026-09-25, /populate) and in WWT-PRC-001 v0.2, Key design choices.

■ Decision

Table 1. Decided items.

#	Item	Decision
D1	Chlorine sensor	Option (a): a membrane-free graphite three-electrode sensor with a potentiostat front end, about \$25, with a monthly DPD comparison built into the maintenance routine. Decided by Amish, 2026-09-25: go with recommendation.
D2	pH	Option (a): site pH entered at installation and checked monthly, no added cost. A pH probe (option b) is kept as a documented variant for sites whose pH varies. <code>budget_usd</code> stays at \$300. Decided by Amish, 2026-09-25: go with recommendation.
D3	Connectivity	Cellular LTE-M and NB-IoT with 2G fallback and SMS alerts for the first build, with the modem as a swappable module so a LoRaWAN module is a variant. This keeps the "GSM or LoRa" pitch. Decided by Amish, 2026-09-25: go with recommendation.
D4	Mounting	A separate pole beside the tapstand, not on it. Decided by Amish, 2026-09-25: go with recommendation.
D5	Sampling	An hourly 90 s timed flush through a latching valve, not continuous flow. Decided by Amish, 2026-09-25: go with recommendation.
D6	Default alert thresholds	Free chlorine below 0.2 mg/L and turbidity above 5 NTU, confirmed by two consecutive readings, adjustable per site. Decided by Amish, 2026-09-25: go with recommendation.
D7	Drain destination	The tapstand basin by default; a soakaway or a container for non-drinking use where users prefer, decided per site. Decided by Amish, 2026-09-25: go with recommendation.
D8	First site type	Tapstands first; tank outlets and kiosks later. Decided by Amish, 2026-09-25: go with recommendation.
D9	Battery	LiFePO4, with charging blocked below 0 °C and above 45 °C, rather than lithium-ion 18650 cells. Decided by Amish, 2026-09-25: go with recommendation.
D10	SwapCell	Not used. WaterWatch needs about 0.5 Wh per day and a 48 V SwapCell pack does not fit. The SwapCell interface v0.3 items therefore do not apply. Decided by Amish, 2026-09-25: go with recommendation.

Table 2. Items that remain open.

#	Item	Status
O1	First co-design partner and region. The TRL 2 note suggested approaching a maintenance service provider, but Amish directed that community designs pick co-design partners per area later, so no partner, partner type or region is chosen here.	Proposed, awaiting Amish
O2	Alert language, recipients and wording beyond the R13 rule; to come from caretakers and operators in co-design. No recommendation was made.	Proposed, awaiting Amish
O3	Who owns the data and who may see it (co-design checklist). No recommendation was made.	Proposed, awaiting Amish

■ Consequences

- `project.yaml` : unchanged apart from the TRL fields. No budget, pitch or problem change was recommended, so `budget_usd` stays at \$300 and the pitch still reads "alerts over GSM or LoRa".

- WWT-PRB-001, WWT-PRC-001 and WWT-REQ-001 are revised to v0.3. The key design choices in the precis are no longer "proposed". No requirement target changes as a result of these decisions; the 5 NTU and 0.2 mg/L defaults in R5 are now decided rather than proposed.
- The TRL 3 calculations (WWT-CAL-001) led to four design changes within these decisions: a smaller flow cell, a pressure-compensating flow regulator, a 30 s settling wait and a ventilated sun shield. They add \$12 to the parts cost, now \$282.
- WWT-CAL-001 v0.1 showed that R1 is not met with D2's site pH at pH 7.5 and above. The relaxed R1 target and the trigger for the pH probe variant that were then proposed are now decided by Amish, 2026-09-25: go with recommendation (WWT-DDR-002). D2 stands for sites at or below pH 7.5 whose pH stays within 0.2 between visits.
- TRL 4 is on hold by Amish's instruction. Nothing in this record authorizes building or testing.